

KEVIN DSOUZA

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SUMMARY

Application Security Engineer with 5+ years of experience in manual penetration testing, source code review, API security, and DevSecOps across enterprise environments. Skilled in Web, API and Mobile assessments, SAST/DAST/SCA triage, CI/CD security integration, and risk-based vulnerability remediation using tools such as Burp Suite, Checkmarx, Semgrep, Snyk, Qualys, and Jenkins. Experienced in working with Java and Node.js applications, conducting deep technical assessments, and partnering with development teams to improve software security posture across complex, large-scale, business-critical enterprise applications.

EXPERIENCE

CITIBANK

1401 W Commercial Blvd, Fort Lauderdale, FL 33309

Information Tech Lead Analyst [VP], Team 6

02/06/2023 – Present

- Performed deep manual source code reviews of Java/Spring services and APIs, identifying authorization flaws, injection and deserialization risks, business logic issues, and misuse of security controls across distributed production systems.
- Conducted deep-dive white-box penetration testing across web, mobile, and API services to validate secure design assumptions and uncover complex attack chains (SSRF, privilege escalation, auth bypass, race conditions).
- Authored high-quality application security assessment reports with reproducible PoCs, risk prioritization, and developer-focused guidance, improving security posture across APIs, auth flows, microservice boundaries, and CI/CD-integrated deployments.
- Validated, triaged, and contextualized SAST and DAST findings by correlating results with code paths and API behavior, reducing false positives and partnering with engineers to deliver secure, production-ready remediations at scale.
- Established risk-based vulnerability prioritization by correlating exploitability, application exposure, and data sensitivity, helping engineering teams focus remediation effort on findings with real-world impact rather than scanner severity alone.
- Conducted secure design reviews and threat modeling for Java/Spring applications, APIs, and authentication workflows, proactively identifying architectural security risks early in the SDLC and guiding engineering teams toward secure-by-design implementations.
- Developed custom Burp Suite tooling and workflows to automate repetitive testing patterns and accelerate vulnerability discovery.
- Developed SAGE, a GitHub-based security search automation tool to proactively identify vulnerable code patterns and recurring security anti-patterns across repositories, accelerating triage and enabling early remediation and reducing recurring security issues.

NORTHEASTERN UNIVERSITY (INFORMATION TECHNOLOGY SERVICES DEPARTMENT)

360 Huntington Avenue, Boston, Massachusetts 02115

Security Research Student (Volunteer unpaid experience as part of Capstone Project, Not employment)

09/15/2022 – 12/07/2022

- Designed and implemented an automated vulnerability discovery framework using open-source tooling to detect OWASP Top 10 application risks, integrating scanning logic, result normalization to help remediation across university-managed web environments.

- Automated discovery of live hosts across distributed subnets and executed targeted application security assessments, integrating scan outputs with centralized logging platforms to enable risk analysis, reporting, and remediation tracking at organizational scale

NORTHEASTERN UNIVERSITY (KHOURY COLLEGE OF COMPUTER SCIENCES)

360 Huntington Avenue, Boston, Massachusetts 02115

Graduate Teaching Assistant, CY 5001: Cybersecurity Technologies, Threats, and Defense

09/11/2022-12/16/2022

- Assisted *Professor Sakib Miaz* in teaching core cybersecurity concepts including computer networks, operating systems, server architectures, and scripting, supporting a class of 60+ students.
- Provided mentorship and technical support on topics such as network security, system hardening, and threat identification, improving student performance and lab completion rates.
- Collaborated with faculty to develop lab exercises and course materials focused on virtualization, containerization, and cloud security fundamentals.

NORTHEASTERN UNIVERSITY

360 Huntington Avenue, Boston, Massachusetts 02115

Intramural Sports Official

04/11/2022-12/03/2022

- Officiated intramural cricket, soccer, volleyball, and badminton games, enforcing rules and ensuring fair play across competitive leagues and recreational student tournaments.
- Made real-time decisions during fast-paced matches, resolved player disputes professionally, and maintained safe, controlled, and sportsmanlike playing environments for participants.
- Managed game flow, scoring, and match timing efficiently, helping intramural events run on schedule and align with university recreation standards and league expectations.

NETBRAIN TECHNOLOGIES

15 Blue Sky Dr, Burlington, MA 01803

Application Security Engineer Coop

01/10/2022 – 08/31/2022

- Performed manual penetration testing of web applications and APIs using Burp Suite and Postman, identifying vulnerabilities across authentication, input validation, and business logic layers.
- Executed and triaged SAST, DAST, and SCA scans using SonarQube, Snyk, Qualys, and Mend, prioritizing exploitable vulnerabilities and reducing false positives across application releases.
- Designed and maintained Jenkins CI/CD pipelines to automate build processes and integrate security testing, enabling continuous validation of application releases through SAST, DAST, and SCA tools.
- Configured and integrated Qualys scans within Jenkins pipelines and standalone environments, performing automated DAST scanning to assess security posture across multiple application deployments.
- Conducted software composition analysis using Mend(WhiteSource), identifying vulnerable third-party components and recommending remediation strategies or compensating controls based on risk and business impact.
- Automated provisioning of Linux backend and Windows frontend virtual machines using Jenkins and PowerShell, enabling scalable, repeatable test environments for validating application releases.
- Managed VMware Horizon environments to support virtualized infrastructure, ensuring reliable setup and execution of security testing across multiple operating systems and application environments.
- Administered Linux and Windows-based testbeds using Jenkins and mRemoteNG, facilitating secure access, orchestration, and maintenance of distributed testing environments for application validation.
- Configured reverse proxies for internal tools and services, improving routing, accessibility, and security controls across development and testing environments.

- Maintained comprehensive documentation in Confluence, including vulnerability reports, remediation steps, security policies, and onboarding procedures for accessing internal tools and security workflows.

NORTHEASTERN UNIVERSITY (KHOURY COLLEGE OF COMPUTER SCIENCES)

360 Huntington Avenue, Boston, Massachusetts 02115

Graduate Teaching Assistant for CY5150: Network Security Practices

09/08/2021-12/16/2021

- Assisted in teaching network security concepts including firewalls, VPNs, wireless security, malware threats, and Internet security principles through lectures, case studies, and labs.
- Supported students in understanding common network threats, secure configurations, and defensive technologies, bridging theoretical instruction with real-world cybersecurity applications.
- Guided students through hands-on laboratory exercises, helping them apply network defense concepts and analyze practical security challenges in controlled technical environments.

NORTHEASTERN UNIVERSITY (KHOURY COLLEGE OF COMPUTER SCIENCES)

360 Huntington Avenue, Boston, Massachusetts 02115

Graduate Teaching Assistant for CY5010: Cybersecurity Principles and Practices

06/20/2021-08/25/2021

- Assisted in teaching core information security principles including confidentiality, integrity, and availability (CIA triad), supporting foundational cybersecurity understanding.
- Guided students through security controls, risk management, and incident response procedures, effectively bridging theoretical concepts with hands-on application lab-based exercises.
- Evaluated coursework and assignments to ensure alignment with security best practices and governance frameworks, maintaining academic rigor and reinforcing industry-relevant cybersecurity methodologies.

AZURE SKYNET SOLUTIONS

M-4, First Floor, Old DLF Colony, Sector 14, Gurugram, Haryana, 122001, India.

Cybersecurity Intern

05/01/2020 – 05/30/2020

- Completed a cybersecurity internship focused on security operations and foundational offensive security concepts, gaining practical exposure to vulnerability assessment techniques and real-world security testing.
- Applied vulnerability assessment and security testing methodologies in a professional environment, developing hands-on experience in identifying, analyzing, and understanding common application and system security issues.

SCOUTFLO

201/202 Casablanca, Hiranandani Estate, GB Road, Thane West 400607

Security Intern

06/10/2019 – 04/17/2020

- Assisted in securing and maintaining early-stage cloud and on-prem infrastructure, including basic hardening of Linux and Windows systems across dev/test environments.
- Supported vulnerability identification and remediation by running security scans, reviewing configurations, and assisting in patch management.
- Contributed to access control management — user provisioning, periodic access reviews, and enforcing least-privilege principles.
- Assisted in implementing security best practices during application setup and deployment, including environment configuration and credential handling.

- Participated in log monitoring and basic incident investigation, helping maintain audit trails and compliance-related records. Worked with the team to improve logging and observability for better traceability across systems.

TATA CONSULTANCY SERVICES

Yantra Park, Voltas Compound, Opposite Voltas HRD Centre, Subhash Nagar, Pokhran Road 2, Thane- 400601

Web Development Intern

06/12/2018 – 09/28/2018

- Developed a full-stack job portal application using PHP and MySQL, enabling users to search, filter, and apply for job listings through a dynamic interface.
- Implemented backend functionalities including user authentication, session management, and CRUD operations for job postings, ensuring secure and efficient data handling across the platform.

EDUCATION

Northeastern University [Khoury College of Computer Sciences] | Boston, MA

Master of Science in Cybersecurity

Degree Awarded: December 2022

Don Bosco Institute of Technology (Mumbai University) | Mumbai, India

Bachelor of Technology in Information Technology

Degree Awarded: October 2020

SKILLS & CERTIFICATIONS

Certifications:

- GIAC Certified Web Application Defender (GWEB) (Issued August 2025)
- GIAC Web Application Penetration Tester (GWAPT) (Issued November 2023)
- CompTIA Security+ (Issued October 2021), Expired
- CEH v10 (Issued December 2020), Expired

Core Application Security: Threat Modeling, Secure Code Review, Application Security, Web Application Penetration Testing, API Security Testing, Vulnerability Prioritization (Risk-Based), Defense-in-Depth.

Security Development Lifecycle (SDLC): DevSecOps, CI/CD Security, SAST, DAST, SCA, RASP, Secure SDLC.

Security Tools & Automation: Burp Suite (Custom Tooling), Semgrep, Checkmarx, Snyk, HCL AppScan, Qualys, SonarQube, Mend (WhiteSource), Jenkins, Postman, WAF.

Languages & Frameworks: Java, Spring, Struts, Python (Automation/Scripting), Node.js, Bash/PowerShell.

Foundational & Architecture: REST APIs, OAuth2, JWT, PKI, Docker, Kubernetes, Cloud Security (AWS/Azure).